DSA

Practical 3.4 Quick Sort/ Merge Sort

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Batch:A2

Merge Sort

```
#include<iostream>
using namespace std;

float a[100];

float b[100];

void merge(float a[],int lb,int mid,int ub);

void  mergesort(float a[],int lb,int ub){
      if (lb<ub){
            int mid=(lb+ub)/2;
            mergesort(a, lb, mid);
            mergesort(a,mid+1,ub);
            merge(a,lb,mid,ub);
```

```
    }
}

void merge(float a[],int lb,int mid,int ub){
    int i=lb,j=mid+1,k=lb;
    while(i<=mid && j<=ub){
        if (a[i]<a[j]){
            b[k]=a[i];
            k++,i++;
        }
        else {
            b[k]=a[j];
            k++,j++;
        }
    }
    while (i<=mid){
        b[k]=a[i];
```

```
            k++,i++;
        }
        while (j<=ub){
            b[k]=a[j];
            k++,j++;
        }
        for(i=lb;i<=ub;i++){
            a[i]=b[i];
        }
}

int main (){
        int n;
        cout<<"Enter no of elements in an array:
";
        cin>>n;
        for(int i=0;i<n;i++){
            cout<<"\nEnter "<<i+1<<" Element: ";
```

```
        cin>>a[i];
    }

    mergesort(a,0,n-1);
    cout<<"\nSorted array is: ";
    for (int i=0;i<n;i++){
        cout<<a[i]<<"\t";}
    return 0;
}

/*
Enter no of elements in an array: 5

Enter 1 Element: 58.36

Enter 2 Element: 74.65
```

Enter 3 Element: 74.32

Enter 4 Element: 12

Enter 5 Element: 85.00

Sorted array is: 12 58.36 74.32 74.65 85

*/